Jingbang Chen

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Research Interests

Design, analysis, and implementation of provably efficient graph algorithms and data structures, with extensions on large-scale data management, data mining, machine learning and artificial intelligence.

Career

Research Assistant Professor *Sep 2025 – present*
School of Data Science, The Chinese University of Hong Kong, Shenzhen Shenzhen, China
Research Assistant Professor *Sep 2025 – present*
Center for AI Theoretical Foundation and Systems, Shenzhen Loop Area Institute (SLAI) Shenzhen, China
Co-founder and Co-president *2022 – present*
Universal Cup, https://ucup.ac/

- Co-founded and co-lead a non-profit organization serving the global competitive programming community.
- Provide high-quality training resources and host onsite global events for top competitive programmers.
- Build communication bridges across industry, academia, and the competitive programming community.
- Supported participation from over 8,000 teams from more than 500 affiliations worldwide.
- Hosted the 2024 Universal Cup Summer Summit in Cancun and the 2025 and 2026 Universal Cup Finals in Dongguan and Shanghai.

Education

University of Waterloo *2023 – 2025*
Ph.D. in Computer Science, Algorithms & Complexity Group Waterloo, Canada
Advisor: Richard Peng

Georgia Institute of Technology *2020 – 2022*
M.S. in Computer Science Atlanta, USA
Advisor: Richard Peng

Zhejiang University *2016 – 2020*
B.Eng. (Honors), Pursuit Science Class in Computer Science, Chu Kochen Honors College Hangzhou, China
Advisor: Can Wang

Students

CUHK-Shenzhen

- **Tianran Zhu**, Ph.D., Fall 2026 – current. B.S. Peking University; ICPC Gold; NOI Silver.
- **Junyang Chen**, M.Phil., Fall 2026 – current. B.S. South China Normal University.
- **Jiajun Hou**, M.Phil., Fall 2026 – current. B.S. Beijing Normal-Hong Kong Baptist University.
- **Zixuan Yu**, M.Phil., Fall 2026 – current. B.S. Sun Yat-sen University.
- **Yichun Wang**, M.Phil., Fall 2026 – current. B.S. CUHK-Shenzhen.
- **Xinle Yang**, M.Phil., Fall 2026 – current. B.S. University of Electronic Science and Technology of China.

CUHK-Shenzhen + SLAI

- **Yuyang Zhao**, Ph.D., Fall 2025 – current. B.S. Huaqiao University; ICPC Gold; NOI Silver. Co-advised with Prof. Zhi-Quan (Tom) Luo.
- **Yuanjin Zheng**, Ph.D., Fall 2026 – current. B.S. Zhejiang University.
- **Minghao Li**, Ph.D., Fall 2026 – current. B.S. Nankai University; M.S. The University of Hong Kong.

External Ph.D.

- **Zeyu Wang**, Zhejiang University, Fall 2025 – current. Co-advised with Prof. Can Wang.

Research Assistants

- **Hao Wu**, Zhejiang University, Fall 2025 – current. ICPC Gold.
- **Sichen Wang**, Shenzhen MSU-BIT University, Fall 2025 – current.

- **Jinrong Lai**, Central South University, Fall 2026 – current. ICPC Gold; NOI Silver.
- **Zihao Xiong**, Fall 2026 – current. Undergraduate at Southern University of Science and Technology; NOI Silver.
- **Shicheng Xiao**, Fall 2026 – current. B.S. at Tsinghua University; M.S. at CUHK-Shenzhen; ICPC Gold; NOI Silver.
- **Weinuo Li**, Zhejiang University, Fall 2024 – Spring 2026. B.S. and M.S. Zhejiang University. Now at Yanfu Investments; VLDB 2026; ICDE 2027; ICPC Gold; ICPC Champion; ICPC WF 10th.

Teaching

- CSC 5001, Design and Analysis of Computer Algorithms, CUHK-Shenzhen, Fall 2025. Course / Teacher satisfaction means: 5.90/6.00, 5.94/6.00; responses: 156/200 students.
- CSC 5003, Algorithm Art and Programming Practice, CUHK-Shenzhen, Spring 2026 and Spring 2027 **(Initiated Instructor)**. Spring 2026 Course / Teacher satisfaction means: 6.00/6.00, 6.00/6.00; responses: 46/59 students.
- CSC 3200, Data Structures and Advanced Programming, CUHK-Shenzhen, Fall 2026.
- MF 0005, Analysis of Algorithms, SLAI, Spring 2026 **(Initiated Instructor)**. Co-teaching with Prof. Konstantinos Courcoubetis.

Publications

* indicates corresponding author.

1. *Scalable Algorithm for Dynamic Quasi-clique Detection.*
   Jingbang Chen, Weinuo Li, Yingli Zhou, Hao Wu, Can Wang, Yixiang Fang, Chenhao Ma.
   IEEE International Conference on Data Engineering (ICDE 2027) (CCF-A). arXiv

2. *Scalable Approximate Biclique Counting over Large Bipartite Graphs.*
   Jingbang Chen, Weinuo Li, Yingli Zhou, Hangrui Zhou, Qiuyang Mang, Can Wang, Yixiang Fang, Chenhao Ma.
   International Conference on Very Large Data Bases (VLDB 2026) (CCF-A). arXiv; DOI

3. *Nearly Optimal Internal Dictionary Matching.*
   Jingbang Chen, Jiangqi Dai, Qiuyang Mang, Qingyu Shi, Tingqiang Xu.
   European Symposium on Algorithms (ESA 2026) (CCF-B). arXiv; DOI

4. *Faster Distance Oracles for Dynamic Interval Graphs.*
   Ryan Bai, **Jingbang Chen** (Alphabetical Order), Meng He, Richard Peng, Kaiyu Wu.
   International Symposium on Algorithms and Computation (ISAAC 2026) (CCF-C).

5. *Curing "Miracle Steps" in LLM Math Reasoning with Rubric Rewards.*
   Youliang Yuan, Qiuyang Mang, **Jingbang Chen**, Hong Wan, Xiaoyuan Liu, Junjielong Xu, Jen-tse Huang, Wenxuan Wang, Wenxiang Jiao, Pinjia He.
   Annual Meeting of the Association for Computational Linguistics, Main Conference (ACL 2026) (CCF-A). arXiv

6. *TopKGAT: A Top-K Objective-Driven Architecture for Recommendation.*
   Sirui Chen, Jiawei Chen, Canghong Jin, Sheng Zhou, **Jingbang Chen**, Wujie Sun, Can Wang.
   The Web Conference (WWW 2026) (CCF-A). arXiv; DOI

7. *BEAR: Towards Beam-Search-Aware Optimization for Recommendation with Large Language Models.*
   Weiqin Yang, Bohao Wang, Zhenxiang Xu, Jiawei Chen, Shengjia Zhang, **Jingbang Chen**, Canghong Jin, Can Wang.
   International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026) (CCF-A). arXiv

8. *FrontierCS: Evolving Challenges for Evolving Intelligence.*
   Qiuyang Mang et al., authoring advisor including **Jingbang Chen**, Aleksandra Korolova, Peter Henderson, Pramod Viswanath, Vijay Ganesh, Saining Xie, Zhuang Liu, Dawn Song, Sewon Min, Ion Stoica, Joseph E. Gonzalez, Jingbo Shang, Alvin Cheung.
   International Conference on Machine Learning (ICML 2026) (CCF-A). arXiv

9. *On the Power of Learning-Augmented Search Trees.*
   Jingbang Chen, Xinyuan Cao, Alicia Stepin, Li Chen.
   International Conference on Machine Learning (ICML 2025) (CCF-A). arXiv; Proceedings

10. *Efficient Historical Butterfly Counting in Large Temporal Bipartite Networks via Graph Structure-aware Index.*
Qiuyang Mang, **Jingbang Chen** (Co-first Author), Hangrui Zhou, Yu Gao, Yingli Zhou, Qingyu Shi, Richard Peng, Yixiang Fang, Chenhao Ma.
International Conference on Very Large Data Bases (VLDB 2025) (CCF-A). arXiv; DOI

11. *Scalable Algorithm for Finding Balanced Subgraphs with Tolerance in Signed Networks.*
Jingbang Chen, Qiuyang Mang, Hangrui Zhou, Richard Peng, Yu Gao, Chenhao Ma.
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024) (CCF-A). arXiv; DOI

12. *Distance Queries over Dynamic Interval Graphs.*
Jingbang Chen, Meng He, J. Ian Munro, Richard Peng, Kaiyu Wu, Daniel J. Zhang.
International Symposium on Algorithms and Computation (ISAAC 2023) (CCF-C); Computational Geometry: Theory and Applications, Volume 122 (CCF-C). DOI; Journal DOI

13. *Exponential Convergence of Sinkhorn Under Regularization Scheduling.*
Jingbang Chen, Li Chen, Yang P. Liu, Richard Peng, Arvind Ramaswami.
SIAM Conference on Applied and Computational Discrete Algorithms (ACDA 2023). arXiv; DOI

14. *Hardness of Graph-Structured Algebraic and Symbolic Problems.*
Jingbang Chen, Yu Gao, Yufan Huang, Richard Peng, Runze Wang.
Algorithms and Data Structures Symposium (WADS 2023). arXiv

15. *Fast Adaptively Weighted Matrix Factorization for Recommendation with Implicit Feedback.*
Jiawei Chen, Can Wang, Sheng Zhou, Qihao Shi, **Jingbang Chen**, Yan Feng, Chun Chen.
AAAI Conference on Artificial Intelligence (AAAI 2020) (CCF-A). arXiv; Proceedings

Manuscripts

* indicates corresponding author.

1. *REUF2F: Benchmarking Coding Agents on Formalized Undergraduate Research Problems.*
Zeyu Zheng, Yiyang Liu, Yuyang Zhao, ..., Prasad Tetali, **Jingbang Chen***, Sean Welleck.
Manuscript, 2026. arXiv; GitHub

2. *AdaGEPA: Adaptive Feedback Allocation for Reflective Prompt Optimization.*
Junyang Chen, Zecheng Wang, **Jingbang Chen***.
Manuscript, 2026. arXiv

3. *Scalable Approximate Algorithm for Dynamic Densest Subhypergraphs with Solution-Guided Maintenance.*
Jingbang Chen*, Chenhao Ma, Yingli Zhou.
Manuscript, 2026. arXiv

4. *How Code Helps Different Tasks? A Decompositional Lens on LLM Post-Training.*
Zheng Yu, Yiwei Li, Yishen Chen, Xiang Li, Jiale Han, Benyou Wang, **Jingbang Chen***.
Manuscript, 2026. arXiv

5. *The Price of Order in the Logarithmic Method.*
Sichen Wang, Zhipeng Lu, **Jingbang Chen***.
Manuscript, 2026. arXiv

6. *SkillBloat: Token Amplification Attacks via Skill Injection in LLM Coding Agents.*
Yuanjin Zheng, **Jingbang Chen***.
Manuscript, 2026. arXiv

7. *Finding the Balance Rate of Uncertain Signed Graphs.*
Zeyu Wang, Kudria Sergei, **Jingbang Chen***, Jiawei Chen, Xinyu Wang, Xiaodong Luo, Can Wang.
Manuscript, 2026. arXiv

8. *Sandpile Prediction on Undirected Graphs.*
Ruinian Chang, **Jingbang Chen** (Alphabetical Order), J. Ian Munro, Richard Peng, Qingyu Shi, Zeyu Zheng.
Manuscript, 2023. arXiv

Competitive Programming

Selected results as contestant

- International Grandmaster on Codeforces, max rating 2735.
- Champion, ICPC Asia Qingdao Regional, 2017; Champion, ICPC Southeast USA Regional, 2021; Champion, The IEEEXtreme Programming Competition 15.0, 2021.
- 3rd Place, ICPC North America Championship, 2021; 13th Place, 46th ICPC World Finals, 2024; 56th Place, 42nd ICPC World Finals, 2018.
- 12 gold medals in ICPC/CCPC series contests; Bronze Medal, National Olympiad in Informatics (NOI), 2015.

Coaching, judging, and community

- Coach and deputy leader, Egyptian Olympiad in Informatics (EOI), 2023 – present.
- Trainer, North American Programming Camp (NAPC), 2024 – present.
- Coach, Georgia Tech Programming Team, 2021 – 2023; Student co-coach, Zhejiang University Competitive Programming Team, 2018 – 2021.
- Contest director and chief judge, ICPC National Invitational Contest (Shenzhen), the first ICPC event hosted at CUHK-Shenzhen.
- Chief judge, ICPC East-Continent Finals, 2025.
- Problem setter and judge for ICPC Regionals (Qingdao 2018; Nanjing 2020–2025; Macau 2021–2022; Jinan 2023; Hangzhou 2024), provincial contests, Moscow Pre-Finals Workshop, Petrozavodsk Training Camp, OpenCup, Codeforces Round #573, and Universal Cup.

Service

- Reviewer for WWW 2026, KDD 2026, ISAAC 2025, FOCS 2025, TKDE 2025, KDD 2025, WWW 2025, SODA 2024, SoCG 2024, ICALP 2024, ESA 2024, ACDA 2023.
- Member, SUA Programming Competitions Problems Setter Team.